Core Mathematics

A Practical Guide on Mathematics for Physics Learners

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This document is a set of lecture notes for first-year university students in physics or physical science. The notes cover the mathematical tools most immediately needed: calculus, logic, complex numbers, vectors, and include introductions to vector calculus, differential equations, and statistical analysis. Exercises are central to this document. "Drill" the topics until you can use the tools fast and precisely!

Preface

Physics is an activity to describe the world, but how? At present, mathematics (+ *broken* English) is the only language we can use. So, unfortunately, we need to learn mathematics! (and English...🥺) But, in some sense, you are fortunate, since you have a language. Math and physics have developed together, so you will understand physics better if you study mathematics well!

When you do DIY, you need to use the tools properly, smoothly, and accurately. It is the same here; when we use math as a tool, it must be **logical**, **quick**, and **accurate**. To be logical, you must think each problem carefully; otherwise your discussion might be incomplete and you can't convince others. To be quick and accurate, you must *"drill"* repeatedly, as if top NBA players do shooting practice almost everyday. You should reach the point where basic calculations feel automatic.

This is not a math textbook, so theorem proofs are mostly omitted. **This is rather** an exercise book. Problems and drills are the core of this book.

Target of this document

This document is primarily for first-year undergraduate students in their second semester, who are majoring in physics or physical science.

How to use this book

First, buy a A4-sized paper notebook. You are expected to solve problems, and when you solve "problems", you are strongly recommended to write down your solution process on the notebook, preferably with full descriptions in English. Meanwhile, "drills" are for you to practice calculations, so you can just write down the final answer, or just do it in your brain.

Prepare the textbook [Boas] (see "References" below) and use it as a reference. When you want to understand the details of a topic, or when you want to solve more problems, consult [Boas].

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Visit https://github.com/misho104/LecturePublic for further information, updates, and to report issues.

References

This document is written under the influence of the following references.

- The Math Book by Hal Tasaki (数学：物理を学び楽しむために, in Japanese)

  A free book with full description, but unfortunately in Japanese. Topic selections and the depth of descriptions are based on this reference.

- First-Year Math Practice (in Japanese)

  This document is grounded in a "learn steadily" approach ("じっくり" in Japanese), which originates from Prof. Gocho and Prof. Kiyono, who were instructors of Sho in his freshman. Problems and rigorous math descriptions are taken from their works.

- Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., Wiley, 2006.
- Arfken, Weber, and Harris, *Mathematical Methods for Physicists*, 7th ed., Academic Press, 2023.

  These two books are widely used in physics departments around the world and referred to in this document many times as [Boas] and [AWH]. [Boas] seems more friendly to beginners, while [AWH] seems more detailed and complete. Sho recommends you to study [Boas] *in parallel with* this document, using it as a reference, and to consult [AWH] when you want more details or more exercises.

Lecture Plan

The content is designed for 150-minute × 14-week lectures, as it is originally for a lecture *Mathematics for Fundamental Physics* (基礎物理數學) in National Sun Yat-sen University (國立中山大學).

1. Derivative
2. Units. Significant Digits.
3. Logic. Theorems. a^x and $\log_a x$.
4. Logic. Theorems. a^x and $\log_a x$.
5. Vectors are arrows.
6. $a \cdot b$ and $a \times b$
7. Matrices (rotation, reflection, scaling)
8. Complex numbers (polar form, Euler's formula, multi-valuedness)
9. Complex vectors, Hermitian inner product, bra-ket notation
10. Vector calculus (grad, div, rot; example of point charge)
11. ODE basics
12. ODE basics
13. Probability and error analysis
14. Probability and error analysis

Symbols

Mathematical Notations

Chapter 1	
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	natural numbers (0, 1, 2, ...)
$\mathbb{N}^+$	natural numbers (1, 2, ...)
$\mathbb{R}$	real numbers
$\mathbb{Q}$	rational numbers
$\mathbb{C}$	complex numbers

Greek symbols

A	α	alpha
B	β	beta
Γ	γ	gamma
Δ	δ	delta
E	ε	epsilon
Z	ζ	zeta

H	η	eta
Θ	θ	theta
I	ι	iota
K	κ	kappa
Λ	λ	lambda
M	μ	mu

N	ν	nu
Ξ	ξ	xi
O	o	omicron
Π	π	pi
P	ρ	rho
Σ	σ	sigma

T	τ	tau
Y	υ	upsilon
Φ	φ	phi
X	χ	chi
Ψ	ψ	psi
Ω	ω	omega

Gray-outed ones are almost never used, but others are almost always used. You need to write them so that **others can distinguish each from others**, but Sho thinks Japanese- or Chinese-speakers are OK with it, as we everyday read and write 日 vs 日 vs 白 or 土 vs 士 vs 工 properly!

Quiz

★1. Read out the following words.

(1) α-particle **(2)** β-decay **(3)** Γ-function
(4) ε-δ definition **(5)** angle θ and φ **(6)** μ- and τ-leptons
(7) $\psi(x)$ and $\varphi(x)$ **(8)** ζ and ξ **(9)** metric $\eta_{\mu\nu}$
(10) X-ray and γ-ray **(11)** Ω-baryon **(12)** $\varepsilon_{\mu\nu\rho\sigma}$-tensor

Problems

★0.1. Write the following letters so that people can distinguish each from others.

(1) a, α **(2)** b, β **(3)** c, C **(4)** e, E, ε

(5) ζ, ξ	**(6)** $\Theta, \theta, Q, \sigma, 6$	**(7)** $g, q, 9$	**(8)** $i, l, 1, I$
(9) k, K, κ	**(10)** m, μ	**(11)** p, ρ	**(12)** s, S
(13) t, τ, T	**(14)** u, υ, ν	**(15)** w, W, ω	**(16)** x, X, χ

Chapter 1

Derivative (Review)

1.1 Introduction

> **Remark:** This document is designed for first-year students in their *second semester*, who **already learned derivatives** in their first semester *"calculus"* lecture. If you are not confident in derivatives, you are advised to study the Derivative Boot Camp first[#1], where you should solve (only) the ★ -marked problems there.

Birds sing, fish swim, flowers bloom, stars twinkle, and university students calculate derivatives. So, let's begin with basic derivatives. Do the next quiz! *How many minutes does it take?*

Quiz

★1. Calculate the first derivatives of the following functions.

(1) $(x+1)^3$ **(2)** $\tan x$ **(3)** $2\cos^2 x$ **(4)** $2\cos x^2$

(5) $x^3 \cos x$ **(6)** $\dfrac{3\sin x}{x}$ **(7)** $x^{3/2}$ **(8)** $\sqrt{\sin x}$

If you finished it within two minutes... Wonderful! Most students should take 3–4 minutes. If you took more than four minutes, don't worry. At least you could calculate them, so just a bit more practice will make you perfect. Try some problems in the Derivative Boot Camp

...Oh, you couldn't get the correct answer because of calculation mistakes? That's not a very good sign, as bad as you've forgot how to calculate them. Since you are going to do such calculations more than 100 times in this semester, you should be able to do them fast and precisely.

In university, basic calculations are often overlooked, but their speed and accuracy are important. Since simple calculations appear frequently in physics, speed and accuracy strongly affect your understanding of lectures, your efficiency in self-study, and ultimately your grade. Speed and accuracy can be improved through practice.

This lecture course aims to help you improve such basic calculations, some of which you have already learned in high school, while also introducing new topics that are essential for physics. You are asked to solve many problems and drills, much like children practicing basketball free throws hundreds of times.

[#1] Visit `https://misho104.github.io/LecturePublic` and find `gp1_boot1_deriv_true.pdf`.

1.2 Typical Pitfalls

1.2.1 Notations

Consider a function $f(x)$. The derivative of $f(x)$ is written by $f'(x)$. For example, if $f(x) = 3x^2$, then $f'(x) = 6x$. We may express it in several ways:

$$f'(x) = \frac{\mathrm{d}f}{\mathrm{d}x}(x) = \frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}f(x). \tag{1.1}$$

They all mean the same thing. Don't be confused!

We may consider $f'(x)$ at a specific point, say $x = 3$. If $f(x) = 3x^2$, then $f'(3) = 18$. Typically, we use the following forms to express it:

$$f'(3) = f'(x)\Big|_{x=3} = \frac{\mathrm{d}f}{\mathrm{d}x}(x)\Bigg|_{x=3} = \frac{\mathrm{d}f(x)}{\mathrm{d}x}\Bigg|_{x=3} = \frac{\mathrm{d}f}{\mathrm{d}x}(x)\Bigg|_{x=3} \tag{1.2}$$

but also some authors may write it as $\dfrac{\mathrm{d}f}{\mathrm{d}x}(3)$ or $\dfrac{\mathrm{d}}{\mathrm{d}x}f(3)$.

Students are often confused when we have a **constant** a. If a symbol a is *declared as a constant*, then we may have functions such as $g(x) = ax^2 + a - 3$, for which

$$g'(x) = 2ax. \tag{1.3}$$

(If you are not sure about this, assume $a = 3$. Then $g(x)$ is equal to $f(x)$.)

> **Fail safe note:** If you are puzzled with Equation (1.3), imagine $a = 3$. Then $g(x)$ is equal to $f(x)$, and $g'(x)$ should be equal to $f'(x)$.

Problems

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💪**1.2.** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

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